

SENTINELMIND

CEO-Grade Strategic Blueprint

AI-Powered Behavioral Intelligence
& Early Intervention Platform

Behavioral Digital Twin	Intra-Individual Deviation	B2B2C Universities & Employers
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Executive Summary

"Fitbit for behavioral health — but instead of comparing you to averages, it notices when you stop being you."

Dimension	Detail
What it is	Personalized behavioral baseline per user — monitors deviation, not population comparison.
Core signal	When someone is drifting from themselves, not how they compare to everyone else.
Why it matters	Mental health intervention is reactive, episodic, clinician-dependent. SentinelMind makes it continuous, predictive, and scalable.
Legal positioning	No diagnosis. No detection claims. Identifies behavioral deviations and wellness trends only.
Business model	B2B2C — enterprise (university/employer) pays; individual benefits. Solves willingness-to-pay.

Founder Due Diligence

Biggest Strengths

- Intra-individual methodology is scientifically defensible and clinically meaningful.
- 'No diagnosis' positioning sidesteps FDA Class II/III device regulation — for now.
- University market provides captive population + duty-of-care pressure on buyers.
- Behavioral data compounds into a moat competitors cannot replicate without time.
- B2B2C model — enterprise pays, individual benefits — solves willingness-to-pay.

Biggest Weaknesses

- **Cold-start problem is severe.** Baseline requires 2–4 weeks of consistent data. First-month retention must be solved before the product delivers any value.
- **Data sparsity.** Users will not consistently journal, wear devices, or grant permissions. Signal degrades fast.
- **Clinical validation takes 18–36 months.** You will be selling on vision, not evidence, for years.
- **No clinical team listed.** Red flag for every serious healthcare buyer.
- **iOS passive sensing restrictions.** Post-iOS 14, platform dependency is real and worsening.

Fatal Risks

RISK 1: Liability without clinical moat. If a user deteriorates after the platform flagged 'low risk,' litigation risk is existential. Even with disclaimers, institutional buyers will demand indemnification.

RISK 2: FDA reclassification. FDA 2022 guidance suggests software that 'predicts psychiatric events' may require 510(k) clearance regardless of disclaimers. One high-profile incident triggers this.

RISK 3: Privacy breach with sensitive data. Mental health data breach is not a PR problem — it is company-ending. HIPAA fines + class action + enterprise churn = zero.

RISK 4: Enterprise sales cycle vs. runway. University/health system deals take 9–18 months. If runway is under 24 months at pilot signing, you will die in the valley.

Recommended Pivots If Needed

Trigger	Pivot
Liability risk dominates	Reframe as 'Employee Wellness Analytics' for HR buyers — lower regulatory exposure.

Trigger	Pivot
Clinical validation stalls	Partner with a university research lab as co-investigator — IRB-backed data without funding the study yourself.
Consumer acquisition too expensive	Go pure B2B: sell to self-insured employers first, fund clinical validation from revenue.

Scientific Validation

Claim	Status	Evidence
Sleep disruption predicts mood episodes	PROVEN	Replicated in bipolar, MDD populations (Harvey 2008; Gruber 2011)
Intra-individual deviation > normative benchmarking	PROVEN	Molenaar (2004) idiographic vs. nomothetic; NIMH RDoC framework
Passive sensing detects depression signals	PARTIAL	Saeb et al. (2015) GPS + phone use; requires large N, replication
Journaling NLP predicts mental state change	PARTIAL	LIWC research validated; clinical NLP at scale still nascent
Keystroke dynamics / screen time as biomarker	EXPERIMENTAL	Promising small studies; not replicated at scale
Early intervention reduces crisis escalation	PROVEN	For structured programs (CBT, crisis lines); not yet for AI-triggered
Wearable HRV as anxiety proxy	PARTIAL	Correlation exists; individual variation high; false positive rate underreported

Bottom line: The foundation is solid. The specific combination in an app format at scale is not yet proven. That gap is your clinical validation roadmap — not a blocker.

Market Analysis

Market	Size	Definition
TAM	\$225B	Global mental health market (Grand View Research, 2023)
SAM	\$28B	Digital mental health, employer wellness, university health, EAP modernization
SOM (Y1–Y3)	\$180M	U.S. universities (800+ with counseling mandates), self-insured employers 500–5,000 employees

Best Customer Segments — Ranked

Rank	Segment	Why
1	Universities	Duty-of-care mandate, budget exists (Bipartisan Safer Communities Act), faster procurement than health systems, contained population for research.
2	Self-Insured Employers	Mental health is #1 driver of disability claims. CFO can calculate ROI directly.
3	Health Systems / Payers	Highest contract value; longest sales cycle; requires clinical validation.
4	Consumer	Lowest barrier, highest churn. CAC unsustainable without B2B anchor.

Competitive Analysis & Moat

Competitor Landscape

Competitor	Model	Weakness You Exploit
Wysa	Conversational AI chatbot	Episodic, user-initiated. No passive baseline. No behavioral continuity.
Headspace	Content + guided meditation	Zero clinical signal. No deviation detection. Wellness content only.
Calm	Consumer subscription content	Same as Headspace. Irrelevant in clinical buyer conversations.
BetterHelp	Therapist marketplace	High cost, therapist-dependent, unscalable. No predictive layer.
Talkspace	Therapist marketplace	Under regulatory scrutiny. No behavioral intelligence layer.
Spring Health	EAP + care navigation	Enterprise-focused. No Digital Twin. Navigation, not prevention.
Brightline	Pediatric behavioral health	Narrow population. Telehealth model only.
Koa Health	Employer behavioral wellness	European-focused. Limited U.S. presence. No passive sensing.

White space: None of these do intra-individual deviation detection. That is your moat.

Competitive Moat

Moat Layer	Strength	Time to Build	Notes
Longitudinal behavioral data per user	HIGH	18–36 months	Irreplaceable. Competitors starting today cannot catch up.
Personalized baseline models	HIGH	Ongoing	Each model improves with user data. Cannot be transferred.
Clinical validation partnerships	HIGH	12–24 months	IRB-backed evidence creates procurement credibility.
Integration depth (LMS, EHR, HR)	MEDIUM	6–18 months	Switching cost compounds over time.
Algorithm IP (patents)	MEDIUM	File now	Patentable deviation detection methods.

Product Strategy & MVP Scope

Core Positioning

Not a therapy app. Not a meditation app. A behavioral early warning system that gets smarter the longer someone uses it.

Product Principles

- Passive first — minimize user burden for data collection.
- Never alarm the user — insights framed as 'we noticed something,' not 'you're at risk.'
- Clinician augmentation, not replacement — output goes to counselors, not users directly in clinical settings.
- Privacy by architecture — individual data never exposed to employer/institution.

Product Tiers

Tier	User	Capability
Individual	Consumer / student	Personal behavioral insights, weekly summaries
Professional	Counselor	Aggregate alerts, individual trend access with consent, secure messaging
Enterprise	Institution admin	Population analytics, de-identified trends, care workflow integration

MVP Must-Have

- Onboarding + consent flow (IRB-compliant language, granular per-signal toggles)
- Baseline period: 21 days passive data collection
- Data inputs: mood check-in (2x/day, 10 sec), optional journaling, device activity via HealthKit/Google Fit
- Deviation detection: personal baseline model per user (z-score statistical threshold v1)
- User-facing insight: weekly 'your week vs. your usual' summary — no diagnostic language
- Counselor dashboard: list of flagged users (opted-in), trend view, secure messaging

MVP Explicitly Out of Scope

- Wearable integration (except HealthKit/Google Fit)
- NLP journaling analysis (v2)
- Multi-modal fusion ML (v2)
- EHR integration
- Consumer subscription billing

MVP Success Metric: At a pilot university — 60% of users complete baseline. Counselors report actionable insights in $\geq 30\%$ of flagged cases. Zero privacy incidents.

Functional Requirements

User-Facing

ID	Requirement
F-01	Onboarding with granular consent — passive data, journaling, counselor sharing as separate toggles
F-02	Daily mood check-in: 1–10 scale + optional emoji/word (< 10 seconds)
F-03	Optional free-text journal — E2E encrypted, not accessible to counselor without explicit share
F-04	Passive data ingestion: steps, sleep duration, sleep quality via platform health APIs
F-05	Weekly personal insight report — 'Your baseline vs. this week'
F-06	Opt-in counselor connection with granular data sharing controls
F-07	Push notification for weekly insight (no alarm-style crisis notifications to user)
F-08	Data export and account deletion (GDPR/CCPA compliance)

Counselor Dashboard

ID	Requirement
C-01	Roster view: all opted-in students, sorted by deviation score
C-02	Individual trend view: 30/60/90-day behavioral trajectory
C-03	Alert threshold configuration per institution
C-04	Secure in-app message to user
C-05	Flag documentation: counselor notes attached to each alert
C-06	Aggregate anonymized population report (weekly)

Admin / Platform

ID	Requirement
A-01	Role-based access: user / counselor / institution admin / SentinelMind admin
A-02	Audit log: every data access event logged immutably
A-03	Data residency controls per institution
A-04	BAA execution workflow for HIPAA-covered entities

ID	Requirement
A-05	Consent version tracking — re-consent if policy changes

AI Architecture

V1 — MVP (Statistical)

- Baseline model: Rolling z-score per user per signal (mood, sleep, activity)
- Deviation score: Weighted composite of per-signal z-scores; configurable weights
- Alert threshold: Configurable per institution (default: 2σ deviation sustained 3+ days)
- Rationale: Interpretable, auditable, no black-box liability, fast to build

V2 — Post-Validation (ML)

- Model type: LSTM or Transformer-based time series per user
- Training: Federated learning — model improves without raw data leaving device
- Multi-modal fusion: Sleep + mood + activity → unified behavioral state vector
- NLP journaling: Fine-tuned clinical BERT (MentalBERT / ClinicalBERT) for linguistic markers
- Outcome labeling: IRB study provides ground truth (counselor-confirmed episodes)

V3 — Scale (Causal)

- Causal inference layer: What caused the deviation, not just that it happened
- Intervention recommendation engine: Match deviation pattern to evidence-based micro-interventions
- Population-level pattern detection: Anonymous aggregate signals for institutional early warning

AI Safety Guardrails (Non-Negotiable)

- No output uses the words: diagnose, detect, predict, suicide, crisis — in any user-facing copy
- All ML outputs include uncertainty bounds — never binary 'at risk / not at risk'
- Human-in-the-loop required for all clinical-context alerts
- Model drift detection: alert if baseline model performance degrades

Behavioral Digital Twin Design

Critical design constraint: The Digital Twin model is owned by the user, not the institution. The institution sees only aggregated or consented-shared outputs.

Layer	Components
Raw Data Layer	Mood signals (self-report 2x/day) · Linguistic signals (journal NLP — v2) · Physiological signals (sleep, HRV — wearable v2) · Behavioral signals (activity, screen time where permitted) · Contextual signals (calendar events, semester stress periods)
Baseline Construction Engine	21-day initialization window · Per-signal rolling baseline (mean + variance) · Seasonality adjustment (weekday vs. weekend) · Drift correction (slow legitimate change vs. acute deviation)
Deviation Detection Layer	Per-signal z-score computation · Cross-signal correlation analysis · Temporal pattern analysis (consecutive days deviated) · Composite Behavioral Deviation Index (BDI score 0–100)
Insight Generation Layer	User-facing: plain-language weekly summary · Counselor-facing: ranked alert with signal breakdown · Population-facing: anonymized aggregate trends

Risk Detection & Intervention Framework

Terminology Rule: Never use 'risk detection' in user-facing copy. Use 'wellness trend' or 'pattern change.'

Behavioral Deviation Index (BDI)

BDI Score	Label	Action
0–30	Stable	No action
31–55	Shifting	Weekly insight highlight to user
56–75	Notable change	Counselor soft alert — counselor decides whether to reach out
76–100	Significant deviation	Counselor priority alert + recommended outreach

Intervention Tiers

Tier	Trigger	Action	Owner
1 — Self-care	BDI 31–55	In-app: 'We noticed a shift. Here are some things that tend to help.' (sleep hygiene, breathing, journaling prompts)	System
2 — Soft outreach	BDI 56–75	Counselor dashboard flag; counselor may send check-in message	Counselor
3 — Priority contact	BDI 76–100	Counselor priority alert; recommended direct contact	Counselor
4 — Crisis protocol	Counselor escalation	Warm handoff to institution's existing crisis protocol. SentinelMind does NOT replace 988 or campus crisis line.	Institution

SentinelMind never bypasses the counselor. Tier 4 connects to the institution's existing crisis infrastructure. This is both the ethical position and the liability boundary.

System Architecture

Component	Detail
Client Layer	iOS (Swift/SwiftUI) · Android (Kotlin) · Web (React/Next.js) for counselor dashboard
API Layer	FastAPI (Python) — async; HIPAA-compliant hosting (AWS GovCloud or Azure Government)
Auth Service	OAuth2 + MFA; role-based access control; JWT tokens
Ingestion Service	Passive data collection pipeline; HealthKit + Google Fit adapters
Baseline Engine	Per-user statistical model; Celery workers; nightly recompute
Deviation Engine	Real-time scoring on new data; Redis cache for scoring state
Alert Service	Counselor notification routing; configurable threshold per institution
Audit Service	Immutable append-only event log; every data access recorded
Primary DB	PostgreSQL — user profiles, consent, audit logs
Time-Series DB	TimescaleDB or InfluxDB — behavioral signals
Encrypted Blob	User journal text (AES-256; user-held key in v2)
Cache	Redis — session management, rate limiting, scoring state
Infrastructure	AWS HIPAA-eligible services · VPC isolation per tenant · TLS 1.3 · SOC 2 Type II roadmap from day 1

Security, Privacy & Regulatory Risks

Security & Privacy Controls

Control	Implementation
Data minimization	Collect only signals explicitly consented. No background microphone. No location.
Encryption at rest	AES-256 for all PII. Journal text encrypted with user-derived key (v2).
Encryption in transit	TLS 1.3 minimum across all services.
Zero-knowledge journal (v2)	Journal text never decrypted server-side.
Consent architecture	Granular, revocable, versioned. Separate consent per data type.
HIPAA BAA	Required before any institutional deployment — no exceptions.
FERPA compliance	Critical for universities — student data is covered. Separate architecture required.
Breach response plan	Documented before launch. Not after.
Penetration testing	Annual third-party. Required by enterprise buyers.
SOC 2 Type II	Begin audit evidence at launch. Target certification Month 18.

Regulatory Risk Matrix

Risk	Probability	Impact	Mitigation
FDA reclassification as SaMD	MEDIUM	EXISTENTIAL	Maintain wellness positioning. Retain FDA regulatory counsel. Monitor guidance quarterly.
HIPAA breach	LOW (if built correctly)	EXISTENTIAL	SOC 2 + HIPAA controls from day 1. No shortcuts.
FERPA violation (university)	MEDIUM	HIGH	Separate student data architecture. Legal review before each university contract.
State mental health privacy laws	HIGH	MEDIUM	Privacy law monitoring retainer. Geofenced data handling.
ADA / discrimination claim	LOW	HIGH	Never allow employers to access individual user data under any circumstances.
EU AI Act (if EU expansion)	HIGH in EU	HIGH	Design for transparency and human oversight from day 1.

Business Model

Primary: B2B SaaS

Segment	Pricing	Notes
University	\$8–15 / enrolled student / year	10K-student university = \$80K–\$150K ARR per contract. Target 20 universities in Year 2 = \$1.6M–\$3M ARR.
Self-Insured Employer	\$6–12 / employee / year	1,000-employee company = \$6K–\$12K ARR. Target 50 employers in Year 3 via benefits broker channel.
Health System / Payer	\$25–45 PMPY	Highest contract value. Requires published clinical validation. Year 3–4 target.

Secondary: Clinical Research Data Licensing

- De-identified, aggregated population behavioral data for pharma and academic researchers.
- Requires explicit consent layer + IRB framework.
- Year 3+ only — premature focus here is a distraction.

Unit Economics Target (Year 3)

Metric	University	Employer
CAC (B2B)	\$15,000 per university	\$8,000 per employer
LTV	\$300K (university, 5-year)	\$40K (employer, 3-year)
LTV:CAC	20:1 (university)	5:1 (employer)
Gross Margin	72% target	—

Go-To-Market & University Pilot Plan

Phase	Focus	Actions
Phase 1 M1–12	University Beachhead	3 pilot universities (free/discounted). Entry point: VP Student Affairs or Director of Counseling — NOT IT. Channel: NAMI on Campus, Active Minds.
Phase 2 M12–24	Paid University Expansion	Convert 2 of 3 pilots to paid. Case study + PR. Hire first enterprise AE (higher ed vertical). Target: 15 universities by M24.
Phase 3 M18–30	Employer Market	Partner with 2–3 benefits brokers. SentinelMind pays 15–20% referral on Year 1 contract value. No direct employer sales team yet.
Phase 4 M30+	Health System / Payer	Requires peer-reviewed clinical publication. Partner with health system already in network. Pursue NCQA/URAC accreditation.

University Pilot Blueprint

Item	Detail
Duration	One academic semester (4 months)
Population	200–500 student volunteers (not mandatory). IRB approval required — file jointly with university.
Data collected	Mood, sleep, activity + optional journal
Counselor access	Dashboard for consented students only
Success criteria	65% complete 21-day baseline · Counselors rate insights as 'useful' in ≥30% of flagged cases · Zero FERPA/privacy incidents · Student NPS ≥ 30
Pilot pricing	Free for institution. Students receive \$25 gift card for completing baseline. Budget: ~\$10K per pilot.

Before Approaching Any University — Checklist

- IRB-ready consent documentation
- FERPA legal analysis (written opinion from healthcare attorney)
- HIPAA BAA template reviewed and ready
- Penetration test report or SOC 2 in-progress evidence
- Clinical advisor on letterhead (PhD or MD in behavioral health or psychiatry)

Investor Perspective

Why Invest

- Massive market with genuine unmet need — university mental health is in crisis and buyers are desperate.
- Intra-individual methodology creates real scientific differentiation. This is not a 'me too' wellness app.
- Data moat compounds with time — behavioral longitudinal data is uniquely valuable and cannot be replicated.
- Regulatory positioning is thoughtful — 'wellness' framing buys time to build clinical validation.
- B2B model means enterprise dollars fund clinical validation that then unlocks the health system market.

Why Not Invest

No clinical co-founder. Single biggest red flag. Every sophisticated healthcare buyer will ask who the clinical lead is. Without a PhD/MD in behavioral health on the founding team, credibility is borrowed at best.

Cold-start problem threatens early retention. If users churn before Day 21, the entire product fails to function. This is a product-market fit risk that must be solved with data before Series A.

Long sales cycles + high burn. University and health system deals take 9–18 months. Seed-funded with \$2M and no paying customers = 18 months from the wall.

Liability surface is large. One high-profile adverse event involving a user the platform flagged (or missed) is potentially company-ending, regardless of disclaimers.

iOS data restrictions are worsening. Apple's privacy changes progressively restrict passive sensing. Technical foundation may erode faster than clinical validation progresses.

Patent Opportunities

Opportunity	Strength	Priority	Notes
Intra-individual deviation scoring (composite BDI algorithm)	HIGH	File within 6 months	Novel application of idiographic analysis to multi-signal behavioral data.
Federated learning for personal behavioral baseline without raw data sharing	HIGH	File within 12 months	Privacy + ML combination is patentable and defensible.
Multi-modal behavioral fusion with seasonality correction	MEDIUM	File within 18 months	Evaluate prior art first.
Consent-gated counselor alert routing system	LOW	Evaluate first	Likely prior art exists. Don't file without thorough search.

Recommendation: File provisional patents on BDI algorithm and federated learning approach immediately. Cost: ~\$3K with patent attorney. Blocks competitors and increases acquisition value.

3-Year Roadmap

Period	Milestone	Revenue Target
M1–6	MVP build. IRB approval at 1 university. Clinical advisor hired.	\$0
M6–12	3 university pilots live. Baseline retention data collected. SOC 2 in progress.	\$0–\$50K
M12–18	2 paid university contracts. Seed round closed (\$3–5M). Clinical lead hired.	\$150K–\$400K ARR
M18–24	15 universities. Employer pilot (3–5 companies). First peer-reviewed publication submitted.	\$1.2M–\$2M ARR
M24–30	Series A (\$12–18M). Employer channel launched. Health system pilot begun.	\$3M–\$5M ARR
M30–36	40+ universities. 30+ employers. Clinical validation published. Payer conversations active.	\$7M–\$12M ARR

Team & Hiring Plan

Critical Hires — Pre-Seed / Seed

Role	Priority	Why Critical
Clinical Co-Founder / CCO	HIRE BEFORE FIRST PILOT	Credibility with every buyer. IRB management. Clinical validation design. This hire changes the fundraising story completely.
iOS/Android Engineer	Month 1	Core data collection layer. Must have HealthKit/Google Fit experience.
ML Engineer	Month 3	Time series and behavioral data experience required. Baseline and deviation engine.
Privacy/Healthcare Attorney (fractional)	Month 1	HIPAA, FERPA, consent architecture. Non-negotiable before any pilot.

Post-Seed Hires (Months 12–24)

Role	Notes
Enterprise Account Executive	Higher education vertical experience required
Clinical Research Manager	IRB management, publication pipeline
Security Engineer	SOC 2 Type II execution
Customer Success Manager	Pilot-to-paid conversion focus

Avoid hiring too early: Sales leadership before product-market fit proven = burn without signal.

Final CEO Verdict

"If you were investing your own money, would you invest in SentinelMind, and what would need to happen first?"

VERDICT: Conditionally YES — but not yet. SentinelMind sits at a genuinely important intersection: a real problem (campus mental health crisis), a scientifically defensible approach (intra-individual deviation), a business model that aligns incentives (B2B), and regulatory positioning that buys runway. The data moat is real and market timing is right.

Why I Would Not Write a Check Today — 4 Blockers

BLOCKER 1: No clinical co-founder. Non-negotiable. The company is not credible to hospital systems, universities with sophisticated procurement, IRBs, or Series A healthcare investors without it. This hire changes the story completely.

BLOCKER 2: Cold-start retention is unproven. The product delivers zero value until Day 21. Need evidence — even from a 50-person beta — that real users complete the baseline without a financial incentive. Below 50% completion = fundamental redesign required.

BLOCKER 3: Legal and liability architecture must be bulletproofed. Need written attorney opinion on FDA positioning, completed FERPA analysis, and HIPAA BAA template — before any institutional pilot launches.

BLOCKER 4: Zero paying customers. Even a \$15,000 annual contract changes the fundraising conversation from 'interesting idea' to 'this is real.'

What Would Make Me Invest Immediately

- Clinical co-founder or CCO with relevant credentials announced
- Pilot data showing 60%+ baseline completion rate
- One university signed — paid contract or signed LOI
- Legal architecture documented and reviewed by healthcare attorney

At that point, this is a \$4-\$6M seed at a \$15-\$20M pre-money valuation, targeting 40+ universities and Series A within 20 months. The idea is right. The science is defensible. The timing is real. The team gap and unproven retention are the only things standing between SentinelMind and a fundable company.

Build the clinical team. Prove the retention. Then raise.

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